Tarang Narayan

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Education

Vellore Institute of Technology (VIT) — Chennai, India
Bachelor of Technology in Computer Science and Engineering (Data Science) — *2023 – 2027*

- **Relevant Coursework:** Data Mining, Machine Learning, Deep Learning, Programming for Data Science, Data Structures & Algorithms

Delhi Public School (DPS) — Varanasi, India
Class 12, Higher Secondary Education

Projects

AetherMed: In Silico Clinical Trial Simulation Platform | *PyTorch, Neural ODEs, GNNs, RL* — GitHub

- Architected an end-to-end digital twin pipeline in PyTorch to simulate cohort drug responses and clearance rates using MIMIC-IV clinical telemetry and ChEMBL chemical data.
- Engineered hybrid predictive systems integrating compartmental Physics-Informed Neural ODEs with heterogeneous GNNs to model pharmacokinetic dynamics and predict molecular toxicity risks.
- Implemented offline reinforcement learning (CQL) for adaptive dosage optimization alongside conformal prediction intervals, ensuring well-calibrated uncertainty quantification on out-of-distribution profiles.
- Deployed a production-grade Streamlit application featuring an evidence-grounded RAG reporting module, model serialization, automated unit tests, and CI/CD validation.

BlackSwan Grid Resilience Simulator | *Diffusion Models, MAPPO, XGBoost* — GitHub

- Developed an adversarial stress-testing simulation coupling Generative Diffusion Models with decentralized Multi-Agent Reinforcement Learning (MAPPO) in OpenAI Gym/PettingZoo environments.
- Synthesized compound tail-risk disruptions using diffusion architectures to generate non-linear, multi-node grid failure states that bypass standard $N-1$ heuristics.
- Trained autonomous substation agents (MAPPO) to execute adaptive, real-time load-shedding and transmission rerouting, significantly cutting power loss across cascading failure regimes.
- Built a real-time telemetry dashboard in Streamlit backed by Docker containers, integrating XGBoost risk predictors and quantitative VaR/CVaR metrics to benchmark grid collapse thresholds.

Economic Vitality Index: Spatial Imagery Mining | *R, terra/raster, ML* — GitHub

- Built an automated geospatial computer vision ETL pipeline in R (terra, raster, magick) converting high-resolution satellite imagery into structured spatial-spectral feature matrices.
- Engineered multi-channel signal extractors measuring spectral reflectance shifts, pixel gradient dispersion, and gray-level texture variance to quantify regional infrastructure damage.
- Trained and benchmarked baseline models (Logistic Regression, Random Forest, RBF-Kernel SVM, and MLPs) using spatial cross-validation, achieving an optimal test AUC of 0.85–0.98.
- Synthesized model inference outputs into a regional index, demonstrating strong empirical correlation with macroeconomic proxies (-0.818 against currency devaluation; +0.795 against equity market activity).

Technical Skills

Languages: Python, R, C++, SQL, Bash/Shell
Core ML & Deep Learning: PyTorch, Scikit-learn, XGBoost, Keras, caret, e1071
Specialized ML & Architectures: Heterogeneous Graph Neural Networks (GNNs), Neural ODEs, Generative Diffusion Models, Multi-Agent RL (MAPPO), Offline RL (CQL), RAG, MLPs
Data & Spatial Engineering: NumPy, Pandas, SciPy, Tidyverse (dplyr), terra, raster, magick, Feature Extraction Pipelines
MLOps, Deployment & Tools: Docker, Streamlit, Git/GitHub, CI/CD